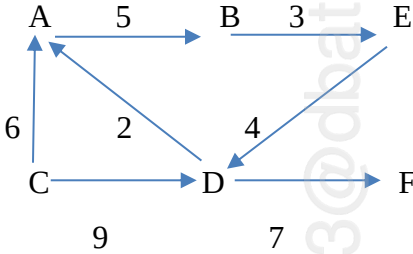


**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****Supplementary Summer Examination – 2024****Course: B. Tech    Branch: Computer Engineering/Computer Science and Engineering    Sem: III****Subject Name: Data Structures****Subject Code: BTCOC303****Max Marks: 60****Date: - 04/07/2024****Duration: - 3 Hrs.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question / answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Assume data wherever necessary and mention it clearly.

|             |                                                                                                                                                                                                                                                                                                                               | (Level / CO) | Marks    |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|
| <b>Q. 1</b> | <b>Solve any Two of the following</b>                                                                                                                                                                                                                                                                                         |              |          |
| 1.          | What is primitive data structure? Differentiate between primitive and non-primitive data structures.                                                                                                                                                                                                                          | Understand   | <b>6</b> |
| 2.          | Define space complexity of an algorithm. Determine the space complexity of the following program segment:<br><pre>int sum(int a[ ], int n) {     int x = 0;     for (int i=0; i&lt;n; i++)     {         x = x + a[i];     }     return x; }</pre> Note: Assume integer variable takes 4 bytes of memory space.               | Analyze      | <b>6</b> |
| 3.          | Define collision. Consider a hash table of size 13 and the hash function given is $h(x) = x \bmod 13$ . Demonstrate how Linear probing handles collision while inserting keys 18, 41, 22, 44, 59, 32, 31, 73, in this order.                                                                                                  | Apply        | <b>6</b> |
| <b>Q.2</b>  | <b>Solve any Two of the following</b>                                                                                                                                                                                                                                                                                         |              |          |
| 1.          | Consider the stack of size 5 memory cells. Suppose initially stack contains A, B, C, D (Top of stack). Then the following operations are executed in order. Show the stack top and any other situation raised while doing each of the operations.<br>i) Push(E) ii) Pop(top) iii) Push(F) iv) Push(G) v) Pop(top) vi) Push(H) | Apply        | <b>6</b> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                       |            |   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---|
| 2.         | Write algorithms for enqueue and dequeue operations on circular queue with example.                                                                                                                                                                                                                                                                                                                   | Evaluate   | 6 |
| 3.         | Explain with suitable example concept of binary recursion.                                                                                                                                                                                                                                                                                                                                            | Apply      | 6 |
| <b>Q.3</b> | <b>Solve any Two of the following</b>                                                                                                                                                                                                                                                                                                                                                                 |            |   |
| 1.         | Describe circular linked list operations in detail.                                                                                                                                                                                                                                                                                                                                                   | Evaluate   | 6 |
| 2.         | What is garbage collection? Who will run garbage collection program?                                                                                                                                                                                                                                                                                                                                  | Remember   | 6 |
| 3.         | Justify a linked list is a data structure that is based on dynamic memory allocation. List the advantages of linked list dynamic memory allocation.                                                                                                                                                                                                                                                   | Understand | 6 |
| <b>Q.4</b> | <b>Solve any Two of the following</b>                                                                                                                                                                                                                                                                                                                                                                 |            |   |
| 1.         | Explain different binary search tree traversal methods.                                                                                                                                                                                                                                                                                                                                               | Understand | 6 |
| 2.         | How does graph can be represented using adjacency matrix? What will be the adjacency matrix for the below given directed weighted graph?<br><br> <pre> graph LR     A -- 5 --&gt; B     B -- 3 --&gt; E     C -- 6 --&gt; A     C -- 9 --&gt; D     D -- 2 --&gt; A     D -- 7 --&gt; F     E -- 4 --&gt; D </pre> | Apply      | 6 |
| 3.         | Write short note on balanced trees.                                                                                                                                                                                                                                                                                                                                                                   | Understand | 6 |
| <b>Q.5</b> | <b>Solve any Two of the following</b>                                                                                                                                                                                                                                                                                                                                                                 |            |   |
| 1.         | Describe how dictionaries can be implemented using skip lists.                                                                                                                                                                                                                                                                                                                                        | Evaluate   | 6 |
| 2.         | What is selection sort? Explain in detail with suitable example.                                                                                                                                                                                                                                                                                                                                      | Apply      | 6 |
| 3.         | Write short note on file handling.                                                                                                                                                                                                                                                                                                                                                                    | Understand | 6 |